



ENGINEERING MATHEMATICS - II

UE20MA151

Dr. SIVASANKARI. V
Department of Science & Humanities

ENGINEERING MATHEMATICS - II



Unit 4 : Inverse Laplace Transform

Session : 8

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$$1) L^{-1} \left\{ \frac{1}{(s-1)^5(s+2)} \right\}$$

Solution:

$$\text{Let } F(s) = \frac{1}{(s-1)^5} \text{ and } G(s) = \frac{1}{s+2}.$$

$$\text{Then } L^{-1} \left[\frac{1}{(s-1)^5} \right] = e^t \frac{t^4}{4!} = f(t) \text{ and}$$

$$L^{-1} \left[\frac{1}{s+2} \right] = e^{-2t} = g(t).$$

Using Convolution Theorem,

$$\begin{aligned} L^{-1}[F(s).G(s)] &= f(t) * g(t) \\ &= \int_0^t f(u)g(t-u)du \end{aligned}$$

$$\begin{aligned} L^{-1} \left\{ \frac{1}{(s-1)^5(s+2)} \right\} &= \int_0^t f(u)g(t-u)du \\ &= \int_0^t e^u \frac{u^4}{4!} e^{-2(t-u)} du \\ &= \frac{e^{-2t}}{4!} \int_0^t e^{3u} u^4 du \\ &= \frac{e^{-2t}}{4!} \left[u^4 \frac{e^{3u}}{3} - 4 \cdot \frac{u^3}{3} \frac{e^{3u}}{3} + \frac{12u^2}{9} \frac{e^{3u}}{3} - \frac{24u}{27} \frac{e^{3u}}{3} + \frac{24}{81} \frac{e^{3u}}{3} \right]_0^t \\ &= \frac{e^t}{24} \left[\frac{t^4}{3} - \frac{4}{9} t^3 + \frac{4}{9} t^2 - \frac{8t}{27} + \frac{8}{81} \right] - \frac{e^{-2t}}{243} \end{aligned}$$

$$2) L^{-1} \left\{ \frac{s^2}{(s^2 + a^2)(s^2 + b^2)} \right\}$$

Solution:

$$\text{Let } F(s) = \frac{s}{s^2 + a^2} \text{ and } G(s) = \frac{s}{s^2 + b^2}.$$

$$\text{Then } L^{-1} \left[\frac{s}{s^2 + a^2} \right] = \cos at = f(t) \text{ and } L^{-1} \left[\frac{s}{s^2 + b^2} \right] = \cos b t = g(t).$$

Using Convolution Theorem,

$$\begin{aligned} L^{-1}[F(s) \cdot G(s)] &= f(t) * g(t) \\ &= \int_0^t f(u)g(t-u)du \end{aligned}$$

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INVERSE LAPLACE TRANSFORM



$$L^{-1} \left\{ \frac{s^2}{(s^2+a^2)(s^2+b^2)} \right\} = \int_0^t f(u)g(t-u)du$$

$$= \int_0^t \cos au \cos b(t-u) du$$

$$= \frac{1}{2} \int_0^t \cos[(a-b)u + bt] + \cos[(a+b)u - bt] du$$

$$= \frac{1}{2} \left[\frac{\sin[(a-b)u + bt]}{a-b} + \frac{\sin[(a+b)u - bt]}{a+b} \right]_0^t$$

$$= \frac{1}{2} \left[\frac{\sin at - \sin bt}{a-b} + \frac{\sin at + \sin bt}{a+b} \right]$$

$$L^{-1} \left\{ \frac{s^2}{(s^2+a^2)(s^2+b^2)} \right\} = \frac{a \sin at - b \sin bt}{a^2 - b^2}$$



THANK YOU

Dr. SIVASANKARI. V

Department of Science & Humanities

sivasankariv@pes.edu